

ALCHEMY AOE ALLIANCE

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MTHD--006

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Prize Distribution, Round Robin

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1 - Definitions

The following mathematical definitions apply to player statistics within the same Round-Robin tournament:

- G – Tournament Grade. A dimensionless value on the scale of 1, used to reflect how well players performed in the tournament. Unless otherwise specified by handbook or equivalent, Grades used to select winning players are repeated here. For transparency, Tournament Grade can never be raised to any power other than 1.
- s – Skill Factor. A value from 0 to 1, calculated as the quotient of a player's tournament elo and the average tournament elo that was calculated among players of their division.
- γ – Skill Bias. A value defined by tournament host to weigh the importance of Skill Factor s over other scoring criteria. Typically exceeds 1 and is used to attract exceptional players who are presumed to invent/inspire effective and entertaining strategies. Equal to 1.25 unless otherwise specified by handbook or equivalent.

2 - Overview

This document specifies a method for distributing prizes in a round robin style tournament, factoring in performance factor and elo distribution.

3 - Background

The original document describing player compensation, MTHD-001, has been divided into two parts and is now deprecated. The second (this specification) defines how winning players are compensated.



4 - Formulation

4.A - Personal Contribution

Personal Tournament Contribution f is a combination of the variables defined in §1, satisfying Equation 4.A below:

$$f = G(p)^\alpha$$

The relationship of any one winning player's Personal Tournament Contribution to those of other winning players determines their fraction of the prize pool.

4.A.1 - Example

A Round-Robin player was selected to receive a prize in a Round Robin tournament after achieving a Tournament Grade of 1.2. Calculate Personal Tournament Contribution if that player had tournament elo of 1267, and the average in the tournament was 1513, with skill bias of 1.5.

From the problem statement, variables are identified:

$$\text{Tournament Grade (} G \text{)} = 1.2$$

$$\text{Skill Factor (} s \text{)} = \frac{1267}{1513}$$

$$\text{Skill Bias (} \gamma \text{)} = 1.5$$

The remaining step is to calculate Personal Tournament Contribution:

$$f = G(p)^\alpha = 1.2 \left(\frac{1267}{1513} \right)^{1.5} = 0.9196$$



4.A.2 - Example

Player1 and Player2 are participating in the same Round Robin tournament and have elos 1833 and 1779, respectively. If both perform well enough to earn prizes from the same division, calculate the Tournament Grade Player1 would need to secure the same fraction of winnings, if Player2's Tournament Grade was 1.04.

To solve, each player gets their own version of Equation 4.A:

$$f_1 = G_1(s_1)^y$$

$$f_2 = G_2(s_2)^y$$

To achieve the same Personal Tournament Contribution, $f_1 = f_2$:

$$G_1(s_1)^y = G_2(s_2)^y$$

Rewritten:

$$G_1 = G_2 \frac{(s_2)^y}{(s_1)^y} = G_2 \left(\frac{s_2}{s_1} \right)^y$$

Recognizing that the skill factor is the tournament elo of a player divided by the average of the group ($\bar{E}$):

$$s_1 = \frac{E_1}{\bar{E}} \quad s_2 = \frac{E_2}{\bar{E}}$$

Such that:

$$\frac{s_2}{s_1} = \left(\frac{E_2}{\bar{E}} \right) \left(\frac{\bar{E}}{E_1} \right) = \frac{E_2}{E_1}$$

Average tournament elo does not matter, as long as Players are in the same division of the tournament:

$$G_1 = G_2 \left(\frac{s_2}{s_1} \right)^y = G_2 \left(\frac{E_2}{E_1} \right)^y$$

Since skill bias factors was not mentioned, it is assumed to be 1.25 per §1:



$$\text{Skill Bias (} \gamma \text{)} = 1.25$$

Collecting other variables from the problem statement:

$$\text{Player1 Elo (} E_1 \text{)} = 1883$$

$$\text{Player2 Elo (} E_2 \text{)} = 1779$$

$$\text{Player2 Tournament Grade (} G_2 \text{)} = 1.04$$

Then finally:

$$G_1 = G_2 \left(\frac{E_2}{E_1} \right)^\gamma = (1.04) \left(\frac{1779}{1883} \right)^{1.25} = 0.9687$$

Thus, in order to achieve the same proportion of tournament earnings, Player1 would only require Tournament Grade of 0.9687 while Player 2 has Tournament Grade of 1.04.

4.B - Aggregate Contribution

Aggregate Tournament Contribution F is determined per Equation 4.B:

$$F = \sum_{i=1}^N f_i$$

Where N is the number of players in the division and f_i is the Personal Tournament Contribution for each player, calculated using Equation 4.A.

4.B.1 - Example

A Round-Robin tournament has the following Personal Tournament Contributions for each of its five players: 0.5, 0.6, 0.7, 0.8, 0.9. Calculate the Aggregate Tournament Contribution.

Applying Equation 4.B, it is straightforward to find the sum:

$$F = \sum_{i=1}^N f_i = f_1 + f_2 + f_3 + f_4 + f_5 = 0.5 + 0.6 + 0.7 + 0.8 + 0.9 = 3.5$$



4.C - Prize Fraction

The Prize Fraction awarded to any player is the same as their Contribution Fraction, which is calculated by dividing Aggregate Tournament Contribution (per Equation 4.B) into Personal Tournament Contribution (per Equation 4.A).

The sum of these fractions across all players must equal 1.

4.C.1 - Example

Calculate the prize fraction of the fourth player in Example 4.B.1:

Obtained simply by dividing 3.5 into 0.8, and getting 0.2286.

4.D - Prize

The Prize awarded to any player is the product of their Prize Fraction and the total prize pool for their division.

The sum of prizes across all players must equal the total prize pool.

4.D.1 - Example

Calculate player winnings from Example 4.C.1, if the tournament has total prize pool of \$300 USD.

Obtained by simply multiplying 0.2286 and \$300 USD, obtaining \$68.57 USD.

5 - Conclusion

A model was presented for compensating winners of a round-robin style tournament that accounts for their performance (Tournament Grade) and skill level (elo), with the option to adjust importance of tournament elo (skill bias).

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